Counterfactual Training

Teaching Models Plausible and Actionable Explanations

Patrick Altmeyer Aleksander Buszydlik Arie van Deursen
Cynthia C. S. Liem

Delft University of Technology

2026-03-23

Training Opaque Models

Tweaking Parameters

Objective:

$$\min_{\theta}\{\text{yloss}(M_\theta(\mathbf{x}), \mathbf{y})\}$$

Training Opaque Models

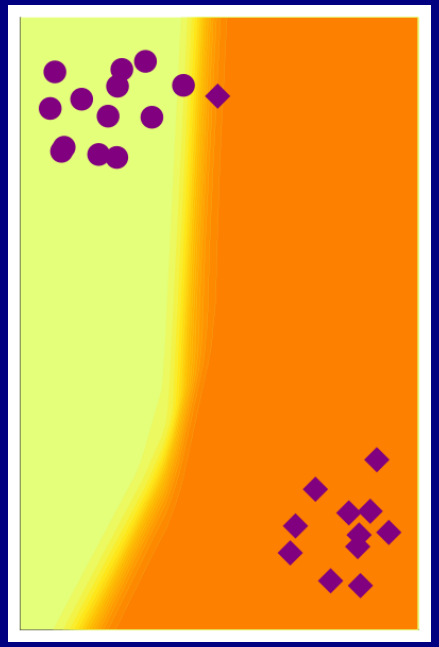
Tweaking Parameters

Objective:

$$\min_{\theta}\{\text{yloss}(M_\theta(\mathbf{x}), \mathbf{y})\}$$

Solution:

$$\theta_{t+1} = \theta_t - \nabla_\theta\{\text{yloss}(M_\theta(\mathbf{x}), \mathbf{y})\}$$
$$\theta^* = \theta_T$$

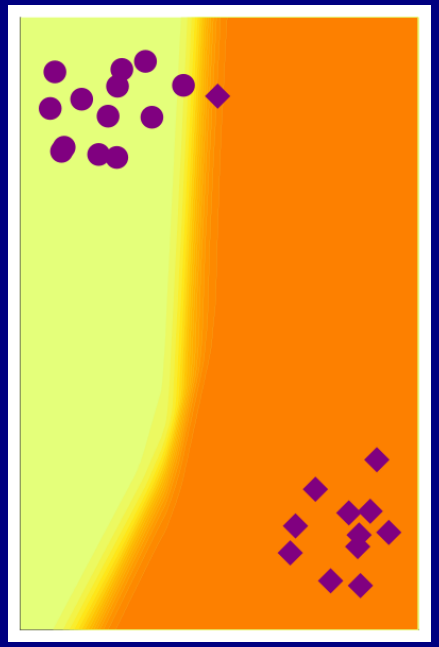


Explaining Opaque Models

Tweaking Inputs

Objective:

$$\min_{\mathbf{x}}\{\text{yloss}(M_{\theta^*}(\mathbf{x}), \mathbf{y}^+) + \lambda \text{reg}(\mathbf{x}, \cdot)\}$$



Explaining Opaque Models

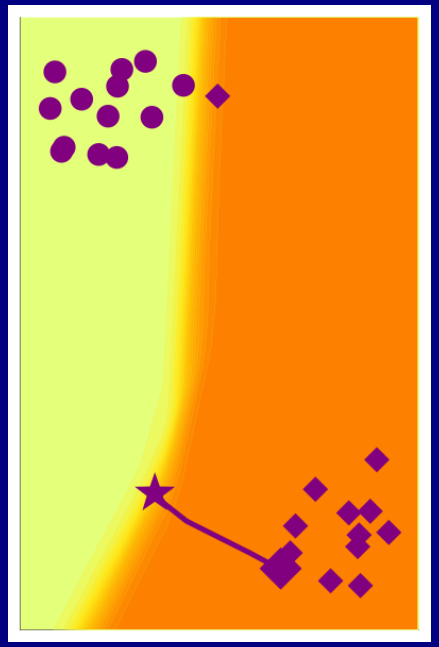
Tweaking Inputs

Objective:

$$\min_{\mathbf{x}}\{\text{yloss}(M_{\theta^*}(\mathbf{x}), \mathbf{y}^+) + \lambda \text{reg}(\mathbf{x}, \cdot)\}$$

Solution:

$$\begin{aligned}\mathbf{x}_{t+1} &= \mathbf{x}_t - \nabla_{\mathbf{x}}\{\text{yloss}(M_{\theta^*}(\mathbf{x}), \mathbf{y}^+) \\ &\quad + \lambda \text{reg}(\mathbf{x}, \cdot)\} \\ \mathbf{x}^* &= \mathbf{x}_T\end{aligned}$$



Explanation or Adversarial Example?

Plausibility at all cost?

All of these counterfactuals are valid explanations for the model's prediction.

Pick your poison ...

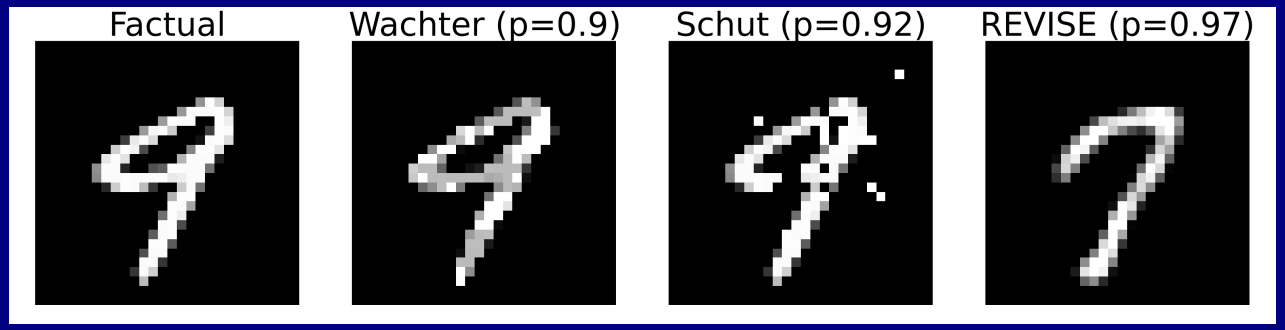


Figure 1: Turning a 9 into a 7: Counterfactual explanations for an image classifier using different approaches (Altmeyer et al. 2024).

Faithful First, Plausible Second

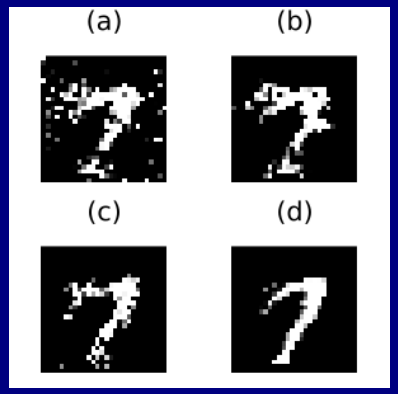


Figure 2: Turning a 9 into a 7. *ECCCo* applied to MLP (a), Ensemble (b), JEM (c), JEM Ensemble (d).

🔑 **Insight**: faithfulness facilitates[1]

- model quality checks (Figure 2).
- state-of-the-art plausibility (Figure 3).

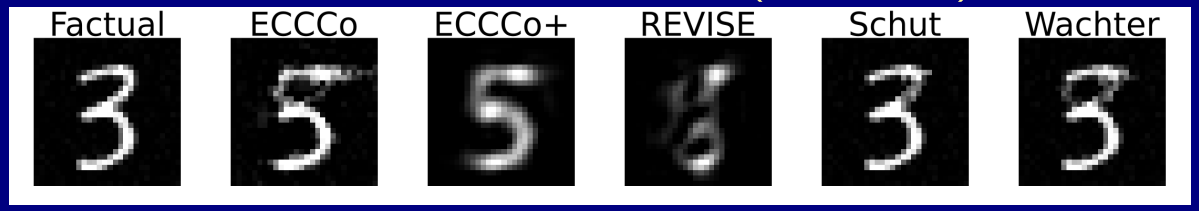


Figure 3: Results for different generators (from 3 to 5).

[1] Altmeyer et al. (2024) @ AAAI 2024. [blog]

Putting it all together

Counterfactual Training

First, *Tweaking Inputs*[2]

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \nabla_\theta \{ECCCo(M_{\theta^*}(\mathbf{x}), \mathbf{y}^+)\}$$
$$\mathbf{x}^* = \mathbf{x}_T$$

Then, *Tweaking Parameters*

$$\theta_{t+1} = \theta_t - \nabla_\theta \{\text{yloss}(M_\theta(\mathbf{x}), \mathbf{y}) + \text{div}(\mathbf{x}^*, \mathbf{x}^+; y^+, \theta)\}$$
$$\theta^* = \theta_T$$

[2] Generate faithful explanations using *ECCCo* objective (Altmeyer et al. 2024).

Counterfactual Training

1. Contrast faithful CE with data $\rightarrow$ **Explainability** $\uparrow$

Counterfactual Training

1. Contrast faithful CE with data $\rightarrow$ **Explainability** $\uparrow$
2. Feature mutability constraints $\rightarrow$ **Actionability** $\uparrow$(holds provably under certain assumptions)

Counterfactual Training

1. Contrast faithful CE with data $\rightarrow$ **Explainability** $\uparrow$
2. Feature mutability constraints $\rightarrow$ **Actionability** $\uparrow$(holds provably under certain assumptions)
3. Bonus: use nascent CE as AE $\rightarrow$ **Robustness** $\uparrow$

Counterfactual Training

1. Contrast faithful CE with data $\rightarrow$ **Explainability** $\uparrow$
2. Feature mutability constraints $\rightarrow$ **Actionability** $\uparrow$(holds provably under certain assumptions)
3. Bonus: use nascent CE as AE $\rightarrow$ **Robustness** $\uparrow$

Counterfactual Training

1. Contrast faithful CE with data → **Explainability** ↑
2. Feature mutability constraints → **Actionability** ↑(holds provably under certain assumptions)
3. Bonus: use nascent CE as AE → **Robustness** ↑

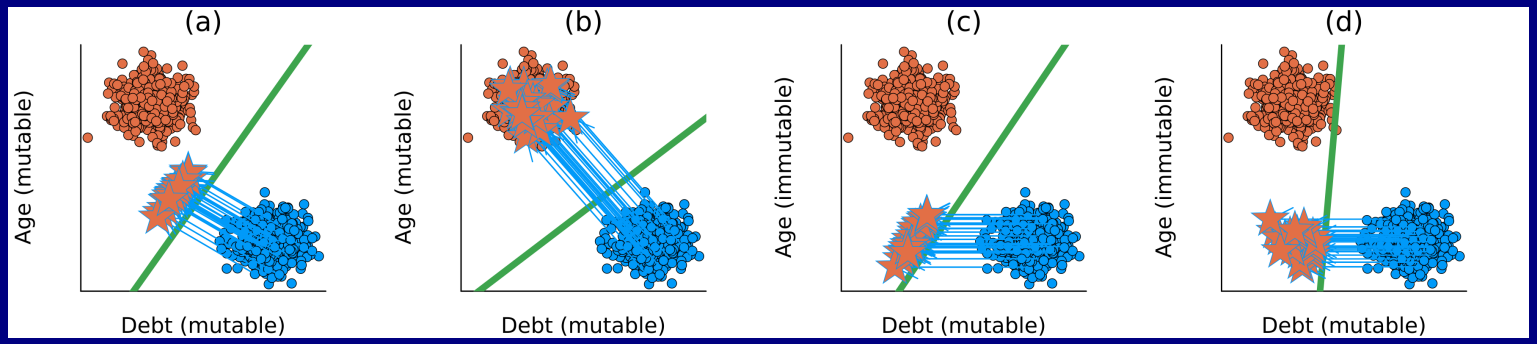


Figure 4: (a) conventional training, all mutable; (b) CT, all mutable; (c) conventional, *age* immutable; (d) CT, *age* immutable.

Counterfactual Training: Results

Figure 5: **Plausibility**: Visual explanations (counterfactuals) for baseline (top row) vs CT (bottom).

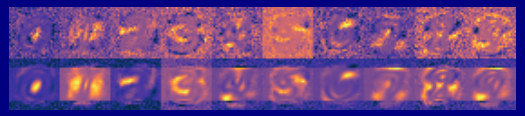


Figure 6: **Actionability**: Visual explanations (integrated gradients) as before. Five top and bottom rows immutable.

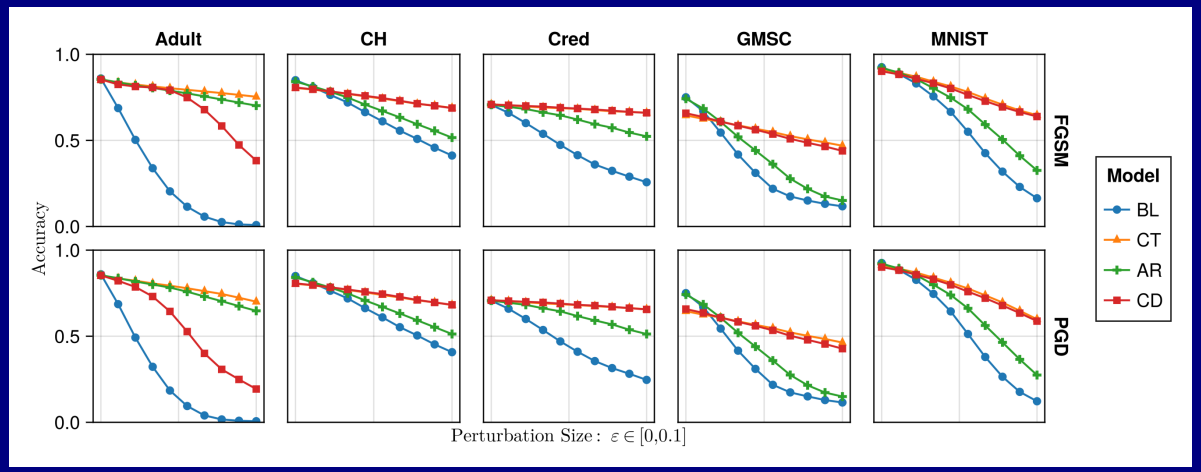


Figure 7: Test accuracies on adversarially perturbed data with varying perturbation sizes.

The Hard Numbers

Extensive experiments and ablation studies on nine datasets–synthetic, tabular and vision–generating millions of counterfactuals:[3]

1. **Plausibility** of CEs increases by up to 90%.
2. **Actionability**: cost of reaching valid counterfactuals with protected features decreases by 19% on average.
3. Models' adversarial **robustness** improves consistently.

[3] Facilitated by our CounterfactualExplanations.jl (Altmeyer, Deursen, and Liem 2023) with multi-processing support and (DHPC) (2022).

Check it out!

Figure 8: Preprint

Figure 9: Software

Figure 10: Homepage

References

Altmeyer, Patrick, Arie van Deursen, and Cynthia C. S. Liem. 2023. "Explaining Black-Box Models through Counterfactuals." In *Proceedings of the JuliaCon Conferences*, 1:130. https://doi.org/10.21105/jcon.00130.

Altmeyer, Patrick, Mojtaba Farmanbar, Arie van Deursen, and Cynthia C. S. Liem. 2024. "Faithful Model Explanations through Energy-Constrained Conformal Counterfactuals." In *Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence*, 38:10829–37. 10. https://doi.org/10.1609/aaai.v38i10.28956.

(DHPC), Delft High Performance Computing Centre. 2022. "DelftBlue Supercomputer (Phase 1)." https://www.tudelft.nl/dhpc/ark:/44463/DelftBluePhase1.